

ADVANCED QUANTUM MECHANICS

Exercise 5 – presentation session – 11.11.2025



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1. Addition of spins and perturbation

Consider two distinguishable spin- $\frac{1}{2}$ particles, labeled A and B , with Hilbert space $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$. Each particle has spin operators $\hat{\mathbf{S}}_A$ and $\hat{\mathbf{S}}_B$ satisfying

$$\hat{S}_{Az} |\uparrow\rangle_A = \frac{\hbar}{2} |\uparrow\rangle_A, \quad \hat{S}_{Az} |\downarrow\rangle_A = -\frac{\hbar}{2} |\downarrow\rangle_A,$$

and similarly for particle B . Define the total spin operator

$$\hat{\mathbf{S}} = \hat{\mathbf{S}}_A + \hat{\mathbf{S}}_B, \quad \hat{S}^2 = (\hat{S}_x)^2 + (\hat{S}_y)^2 + (\hat{S}_z)^2.$$

The unperturbed Hamiltonian is

$$\hat{H}_0 = \mu_B B (\hat{S}_{Az} + \hat{S}_{Bz}),$$

and a weak coupling between the spins is described by the perturbation

$$\hat{V} = J \hat{\mathbf{S}}_A \cdot \hat{\mathbf{S}}_B, \quad |J| \ll 1.$$

(a) Write the product basis

$$\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\},$$

and compute their unperturbed energies under $\hat{H}_0$.

(b) Define

$$|1, 1\rangle = |\uparrow\uparrow\rangle, \quad |1, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), \quad |1, -1\rangle = |\downarrow\downarrow\rangle, \quad |0, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle),$$

and verify that these are eigenstates of $\hat{S}^2$ with eigenvalues $S(S+1)\hbar^2$ for $S = 1, 0$.

(c) Using the identity

$$\hat{\mathbf{S}}_A \cdot \hat{\mathbf{S}}_B = \frac{1}{2}(\hat{S}^2 - \hat{S}_A^2 - \hat{S}_B^2),$$

compute

$$E_S^{(1)} = J \langle S, M | \hat{\mathbf{S}}_A \cdot \hat{\mathbf{S}}_B | S, M \rangle$$

for the singlet ($S = 0$) and triplet ($S = 1$) states.

(d) Determine the energy splitting between the triplet and singlet manifolds and sketch the spectrum for $J > 0$ (antiferromagnetic) and $J < 0$ (ferromagnetic) couplings.

2. Time independent perturbation theory

Let us consider a quantum harmonic oscillator

$$\hat{H}_0 = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2 \hat{x}^2, \quad \hat{H}_0 |n\rangle = E_n^{(0)} |n\rangle, \quad E_n^{(0)} = \hbar\omega \left(n + \frac{1}{2}\right).$$

with

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle, \quad \hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle, \quad \hat{x} = x_0(\hat{a} + \hat{a}^\dagger) = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger).$$

Take a weak quartic anharmonicity

$$\hat{V} = \lambda \hat{x}^4, \quad |\lambda| \ll 1.$$

(a) Compute

$$E_n^{(1)} = \langle n | \hat{V} | n \rangle = \lambda \langle n | \hat{x}^4 | n \rangle.$$

(b) Show that

$$\langle k | \hat{x}^4 | n \rangle \neq 0 \quad \text{only for } k = n, n \pm 2, n \pm 4.$$

Derive explicit formulas for the five nonzero matrix elements $\langle n | \hat{x}^4 | n \rangle$, $\langle n \pm 2 | \hat{x}^4 | n \rangle$, and $\langle n \pm 4 | \hat{x}^4 | n \rangle$.

(c) Write

$$E_n^{(2)} = \sum_{k \neq n} \frac{|\langle k | \hat{V} | n \rangle|^2}{E_n^{(0)} - E_k^{(0)}},$$

and reduce the sum to the contributing $k \in \{n \pm 2, n \pm 4\}$ from part (b). Leave the result in closed form in terms of n .

Hint Use the zero point motion x_0 and the perturbative parameter

$$\epsilon = \frac{\lambda x_0^4}{\hbar \omega}.$$

to simplify the calculations and the final results.

3. Fermi's golden rule

We consider the same quantum harmonic oscillator, this time driven by a time-dependent force of the form

$$\hat{V}(t) = F_0 \hat{x} \cos(\Omega t) = \frac{F_0 x_0}{2} (\hat{a} + \hat{a}^\dagger) (e^{i\Omega t} + e^{-i\Omega t}), \quad F_0 x_0 \ll \hbar \omega_0$$

(a) Compute the matrix elements

$$\langle n \pm 1 | \hat{x} | n \rangle = x_0 \sqrt{n+1} \delta_{m,n+1} + x_0 \sqrt{n} \delta_{m,n-1}.$$

(b) Using selection rules from (a), show that the perturbation can only connect $|n\rangle \leftrightarrow |n \pm 1\rangle$, corresponding to absorption (+) and emission (−) of a quantum of the drive.

(c) Using

$$\Gamma_{i \rightarrow f} = \frac{2\pi}{\hbar} |\langle f | \hat{V}(\Omega) | i \rangle|^2 \delta(E_f - E_i \mp \hbar \Omega),$$

write the explicit transition rates $\Gamma_{n \rightarrow n \pm 1}$.

(d) Show that

$$\Gamma_{n \rightarrow n+1} = \frac{\pi F_0^2 x_0^2}{2\hbar^2} (n+1) \delta(\omega_0 - \Omega), \quad \Gamma_{n \rightarrow n-1} = \frac{\pi F_0^2 x_0^2}{2\hbar^2} n \delta(\omega_0 + \Omega),$$

and discuss which process (absorption or emission) is resonant when $\Omega \approx \omega_0$.